CO1 - AT1

Analytical problem Solving

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Sub : Artificial Intelligence

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1. Water Jug problem :-

Given a 4-gallon jug and a 3-gallon jug, obtain exactly 2 gallons in the 4-gallon jug without using any measuring instruments.

Assumptions :

Both jugs start empty

The pump can fill either jug completely.

Water can be poured between jugs or discarded.

There are no measurement markings on the jugs.

Let the state be (x, y), where x is the water in the 4 gallon jug and y is the water in the 3 gallon jug.

Initial state : (0, 0)

Goal state : any state (2, y).

Step	operation	4-gallon jug	3-gallon jug
1.	Fill the 3 gallon jug	0	3
2.	Pour the 3-gallon jug into 4-gallon jug	3	0
3.	Fill the 3 gallon jug again	3	3
4.	Pour into 4-gallon jug until it is full	4	2

5.	Empty the 4-gallon jug onto the ground	0	2
6.	pour the remaining 2 gallons into the 4-gallon jug	2	0

Result: Exactly 2 gallons of water are obtained in the 4-gallon jug.

2. Mars Rover - Intelligent Agent

A mars rover is an intelligent agent that explores mars, collects samples, and sends scientific data to earth.

i) Percepts:

- Images of rocks, soil and terrain
- Temperature, pressure, and obstacle data
- Chemical composition of samples
- Rover position and battery level.

ii) Operating Environment:

- partially observable
- Stochastic
- Sequential
- Dynamic
- Hybrid

iii) Actions:

- Move and turn
- Capture images and scan terrain
- Collect and analyze sample
- Avoid Obstacles
- Send data to Earth and manage battery power.

iv) Performance Evaluvation :

Performance is measured by dample - analysis accuracy, Scientific discoveries, safe navigation, energy efficiency, mision Completion, and quality of transmitted data.

v) Suitable Agent Architecture :

A model-based, goal-based, and utility-based agent is suitable. It maintains an internal model of the envisonment, plan actions to achive Scientific goals.

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3. 8-Queens problem

Problem: place 8 queens on an 8×8 chessboard so that no two queens attack each other.

i) Problem Components :-

Initial state : Empty chessboard

State space : All possible arrangement of queens.

Actions : place a queen in a space position in the next column.

Goal test : All 8 queens are placed without conflicts.

Path cost : Each queen placement - 1

ii) Search stategy :-

Backtracking Search is suitable because it places queens one by one and removes a queen whenever a conflict occurs.

Two queens must not share the dame row, column or diagnol.

iii) Pseudocode :

```
Solve (column) :
    IF column 78 :
        RETURN TRUE
    FOR row = 1 to 8 :
      IF position is SAFE :
          PLACE queen
      REMOVE queen
    RETURN FALSE.
```

iv) One valid Solution :

The row positions for columns 1 to 8 are :

<1, 5, 8, 6, 3, 7, 2, 4>.

Result : All 8 queens are placed safely without attacking one another.

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4. OLA cab booking - Problem solving Agent

Problem: A Customer wants to travel from a source to a destination using OLA and select a suitable cab type.

i) Type of Agent :

A utility - based problem solving agent is suitable because it Compares cab options based on fare, travel, time, Comfort, and availability.

ii) Problem Components :

Initial state : Customer's Current location

Goal state : Reach the destination

Actions : Select cab, choose route, book and travel

Constraints : Budget, Cab availability, and Customes preferences.

iii) Pseudocode :-

```
BEGIN
    INPUT source, destination, budget
     best_cab = NONE
      best_utility = -INFINITY
  FOR each cab type:
     CHECK availability and fare
  IF cab is available AND fare <= budget :
          CALCULATE utility.
     ELSE :
        DISPLAY "NO SUITABLE cab available"
  END.
```

Result : The agent selects the available cab with the highest utility while satisfying the customer's budget.

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5. Uniform Cost Search (UFS)

Problem : Find the minimum-cost path from warehouse S to warehouse G using the weighted graph provided in the assessment.

i) Method selection :

UCS expands the node with the lowest cumulative path cost.

$g(n)$ = Sum of edge costs from S to n.

ii) Step-by-Step Execution :

Step	Node expanded	Cumulative cost
1	S	0
2	A	1
3	C	2
4	D	3
5	B	4

iii) cost calculation :-

The optimal path is :

$$S \rightarrow A \rightarrow C \rightarrow G$$

Total cost :

$$C = 1 + 1 + 2$$
$$= 4 \text{ units.}$$

iv) Result & Interpretation :-

The minimum - cost delivery route is $S \rightarrow A \rightarrow C \rightarrow G$, with a total cost of 4 units.